

EARLY GAME INDICATORS

IN PROFESSIONAL LEAGUE OF LEGENDS

COMP-248 Project in Data Science • Milestone #2

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Progress Report

3

SEASONS

5

REGIONS

4

KEY FINDINGS

PROJECT RECAP

Where we are and what we're solving

DOMAIN

Professional LoL

Esports match data analysis

GOAL

Predictive Power

Which early stats predict wins?

DATA

Oracle's Elixir

3 seasons • 5 main regions

CORE HYPOTHESES

- Which early-game stats are most linked to winning?
- How reliably do early indicators forecast match outcomes?
- Does early-game importance change across seasons?
- Do regions show different early-game patterns?

WORK DONE TO DATE

Progress completed since project proposal

1 Loaded & merged 3 CSV datasets (~366k rows)

2 Cleaned data: team-level filter, removed nulls

3 Created derived features (gold_lead, early_score, etc.)

4 Correlation analysis (gold/xp/cs diff vs result)

5 Year-by-year and region-by-region breakdowns

6 Comeback analysis with regional trends

7 Objective win rate ranking (8 indicators)

8 Combined Early Score (0–5 scale)

9 Game length decay analysis

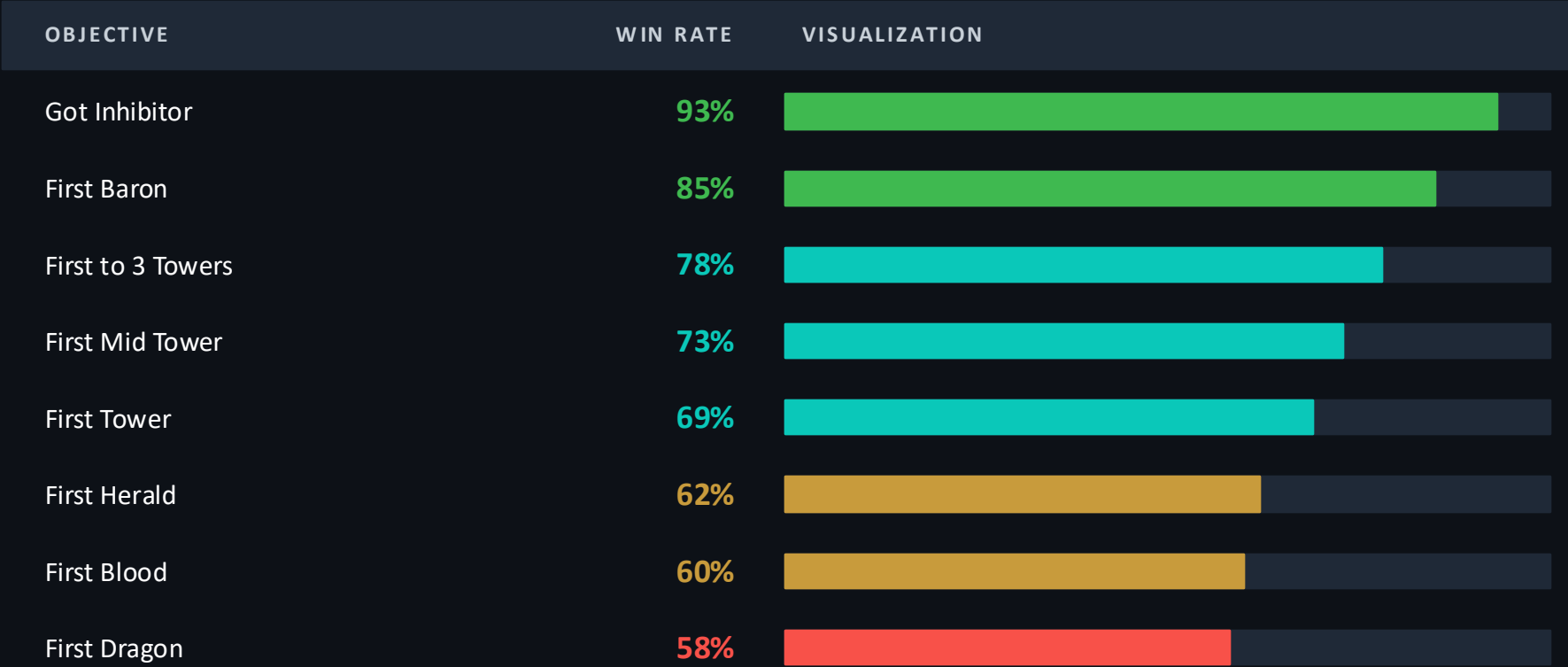
10 False positive analysis (overall, region, year)

11 Causal chain: gold → baron → win

12 Deeper correlations: xp_lead, cs_lead, kd_ratio

FINDING 1: OBJECTIVE WIN RATES

Tower-based objectives outperform kill-based ones



Tower objectives (69-78%) consistently beat first blood (60%) and first dragon (58%)

FINDING 2: COMBINED EARLY SCORE

Win rate scales near-linearly with early dominance

WHAT IS IT?

A score from 0 to 5 counting how many early advantages a team won:

- First Blood
- First Tower
- First Herald
- First Dragon
- Gold Lead at 15

3/5 is the threshold that flips a team from underdog to favorite

0/5



13.5%

1/5



24.2%

2/5



38.8%

3/5



61.3%

4/5



75.8%

5/5



86.5%

FINDING 3: BARON IS THE CONVERSION

Gold lead alone barely wins — baron closes games

Ahead + Got Baron

91.9%

Near unbeatable

Behind + Got Baron

70.7%

Comeback tool

Ahead + No Baron

44.5%

Coin flip

Behind + No Baron

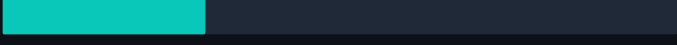
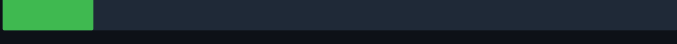
8.6%

Almost lost

Even teams BEHIND can win 70.7% if they steal baron — it's the strongest comeback mechanism

FINDING 4: FALSE POSITIVE RATES

How often does each indicator fail to predict a win?

INDICATOR	FALSE POSITIVE RATE	VISUAL
First Dragon	42.3%	
First Blood	40.5%	
First Herald	38%	
Gold Lead	26.4%	
First Baron	14.9%	
Got Inhibitor	6.6%	

First dragon and first blood fail nearly half the time — least reliable as standalone signals

CHANGES SINCE LAST MEETING

Adjustments based on professor feedback and discoveries

REMOVED

ML Classifier

Per professor's feedback — focus on going deeper into analysis instead of building predictive models

FIXED

Major Data Issue

Discovered 2019 rows were silently dropped due to NaN cascade. Now filtering team-level rows (position == "team")

REFINED

Dataset Filters

Removed leakage from 2020/2023/2026 rows — now strictly limited to 2019, 2022, 2025 seasons

ADDED

Deeper Correlations

XP lead, CS lead, kill/death ratio analysis, and first blood gold impact investigation

WHAT'S NEXT: DEEPER ANALYSIS

Four areas of investigation to strengthen findings

01

Combined Early Score Breakdown

Go deeper into which specific objective combinations make up each score level (0/5 → 5/5) and find which 3-objective combos win most reliably

02

Side Advantage (Blue vs Red)

Does the side a team plays on affect early objective control and overall win rate? Untouched dimension in current analysis

03

Game Length × Objective Decay

Does first baron stay decisive in 40+ minute games or does it lose value like gold leads do? Different objectives may have different shelf lives

04

Patch-Level Drift

Does early-game importance shift on specific patches rather than just year by year? Patches are the actual unit of meta change

FINAL STEPS

Roadmap to project completion

REMAINING DELIVERABLES

KNIME Visualizations

Heatmaps, bar charts, trend lines for results

Notebook Narrative

Markdown documentation between every analysis section

Final Report

Polished PDF + 4 page document if needed

YouTube Video

5-min data storytelling presentation

GitHub Repository

Code + dataset + README pushed publicly

THANK YOU

Questions & Feedback

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